Fiscal Year:	FY 2025	**Task Last Updated:**	FY 03/28/2025
PI Name:	Macias, Brandon Ph.D.		
Project Title:	Mitigating Headward Fluid Shifts with Veno-Constrictive Thigh Cuffs During Spaceflight		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **SANS**:Risk of Spaceflight Associated Neuro-ocular Syndrome (SANS)		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	NASA CENTER	**Phone:**	281-483-2026
Organization Name:	NASA Johnson Space Center		
PI Address 1:	Johnson Space Center Cardiovascular and Vision Laboratory		
PI Address 2:	2101 NASA Parkway, HAC/B21N-1207		
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	36
Comments:	NOTE: Became civil servant fall 2020; previously KBR/NASA Johnson Space Center. Prior to that until 2016, was at the University of California, San Diego.		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	05/15/2021	**End Date:**	05/14/2026
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
Contact Email:	becky.brocato@nasa.gov		
Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:	Additions: Griffin Donley will replace the functional role of John Foulk. Katrina Carter, Jason Lytle, and Sarah Pickering, are new personnel part of the Cardiovascular and Vision Laboratory working on this project. Dr. Alex Huang from UCSD will join the team as Co-I to help with his OCT expertise and past thigh cuff countermeasure research expertise. Departures: John Foulk, Annelise Miller, Chris Miller, Evan Pettit.		
COI Name (Institution):	Laurie, Steven Ph.D. (KBR/NASA Johnson Space Center) Lee, Stuart Ph.D. (KBR/NASA Johnson Space Center) Cole, Chris Ph.D. (Clemson University) Huang, Alex (University of California, San Diego)		
Grant/Contract No.:	Directed Research		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This study will determine the ability of venoconstrictive thigh cuff (VTC) application for varying durations to mitigate the weightlessness-induced headward fluid shift during spaceflight for the purpose of evaluating this device as a countermeasure for SANS. The effects of varying durations of continuous VTC usage (~30 minutes, ~3 hours, and ~6 hours) on ocular and cardiovascular variables will be investigated. This proposed study will help determine the potential efficacy of VTC to mitigate the cephalad fluid shift that is hypothesized to be the primary initiating factor to SANS. The VTC countermeasure technique is low-mass, portable, simple to use and may be used for extended durations while working in an operational spaceflight environment. Specific Aim: 1) Determine the efficacy of VTC application to mitigate a spaceflight-induced headward fluid shift. We hypothesize that a VTC countermeasure will temporarily reverse or attenuate spaceflight-induced ocular and cardiovascular changes.
Rationale for HRP Directed Research:	This research is directed because it contains highly constrained research. This study will determine the ability of veno-constrictive thigh cuffs (VTC) application for varying durations to mitigate the weightlessness-induced headward fluid shift during spaceflight for the purpose of evaluating this device as a countermeasure for Spaceflight Associated Neuro-ocular Syndrome (SANS). The effects of varying durations of continuous VTC usage (30 minutes, 3 hours, and 6 hours) on ocular and cardiovascular variables will be investigated. This proposed study will help determine the potential efficacy of VTC to mitigate the cephalad fluid shift that is hypothesized to be the primary initiating factor to SANS. The VTC countermeasure technique is low-mass, portable, simple to use and may be used for extended durations while working in an operational spaceflight environment. The investigator team has the required expertise to conduct this research and has experience from a ground study led by this team in 33 subjects who wore these VTC for up to ~2 hours while supine at rest. An updated VTC and microfiber sleeve have been developed and tested in these ground-based studies.
Research Impact/Earth Benefits:	The Flight Thigh Cuff research study will advance NASA’s understanding of use of the VTC as a potential countermeasure for Spaceflight Associated Neuro-ocular Syndrome (SANS), an important human health and performance risk. This goal will be accomplished by quantifying vascular and ocular changes before and during spaceflight. Given the unique environment of the International Space Station, commonly used ophthalmic instruments are being implemented in novel operational environments (e.g., optical coherence tomography (OCT) and pneumotonometry). This work will help NASA to evaluate this spaceflight-associated risk more accurately and to refine its data-driven approach to SANS countermeasure development.
Task Progress:	KEY ACCOMPLISHMENTS THIS REPORTING PERIOD Informed Consent • Informed consent briefings were conducted for Crew-11, Crew-12, Boeing commercial flight test, and 75S. • Eight crewmembers have consented to participate in this study which includes an additional five from this reporting period. Preparation Activities • Additional Crew training classes for both operator and subject classes were conducted and completed. Twenty-eight preflight crew training classes have been conducted in this reporting period with overwhelmingly positive feedback from crewmembers. • Flight-certified Venoconstrictive thigh cuffs (VTC) were returned to NASA’s Research Operations and Integration (ROI) group after previous successful data collection aboard the International Space Station (ISS) to undergo biological analysis by the NASA Johnson Space Center (JSC) Microbiology Laboratory. Preliminary results show minimal wear and tear and no concerns of bacterial growth increasing our confidence in the longevity of this potential fluid shift countermeasure. • New hardware was procured to support data collection and improved image quality (9L ultrasound probe and compression sonography device). Scheduling is ongoing for the delivery of this hardware to the ISS for future use. Data Collection Activities • A Test Readiness Review (TRR) was completed on 5/31/2024. • JSC Institutional Review Board (IRB) continuing review approved 10/17/2024. • Three crewmembers have completed all data collection activities. • Four VTC preflight fit checks have been completed. • Four inflight VTC fit checks have been completed. • Three preflight data collections have been completed. • Ten inflight data collections have been completed. Presentations A poster was presented at the 2025 NASA Human Research Program Investigators' Workshop in Galveston, Texas.
Bibliography Type:	Description: (Last Updated: 06/03/2026)
Abstracts for Journals and Proceedings	Poczatek MJ, O’Grady CS, Boothe GS, Laurie SS, Martin DS, Lee SMC, Cole CW, Donley G, Young MH, Macias BR. "Mitigating headward fluid shifts with veno-constrictive thigh cuffs during spaceflight." 2025 NASA Human Research Program Investigators’ Workshop, Galveston, Texas, January 28 -31, 2025. Abstracts. 2025 NASA Human Research Program Investigators’ Workshop, Galveston, Texas, January 28 -31, 2025. , Jan-2025